8.1 When Solutions Aren't Unique

WORKSHEET

The question

STEP III 1994, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself Take the square root of both sides: $y' = \pm 2\sqrt{y}$. Try $y = (x - c)^2$. Does it satisfy the equation? Is there an even simpler solution?

STAGE 1 Solutions through a point

Covers: the two solutions through (a, b^2)

Ask yourself Which parabolas $y = (x - c)^2$ pass through the point (a, b^2)? Solve for c. Why are the two answers different?

STAGE 2 Two points and a common solution

Covers: the points $(a_1, 1)$ and $(a_2, 1)$

Ask yourself Which solution passes through the origin? Where does it reach height 1? Those give you a_1 and a_2. List the two solutions through each point and see which they share.

STAGE 3 The whole family, and a common tangent

Covers: the last part: the common tangent

Pause and try Sketch several of the parabolas $y = (x - c)^2$. Which line touches every one of them? Is that line itself a solution of the equation?

STAGE 4 Why it matters

Covers: the last part: why the tangent is relevant

Ask yourself Which solutions pass through a point on the x-axis? Can you build one that runs along the x-axis for a while and then leaves it along a parabola? Is it differentiable where the pieces join?

What's really going on

When are solutions unique?

The *Picard–Lindelöf theorem* (proved by Émile Picard and Ernst Lindelöf around 1890–1894) says that the equation $y' = F(x, y)$ with $y(x_0) = y_0$ has exactly one solution near x_0, provided F is *Lipschitz* in y: there is a constant L with $|F(x, y_1) - F(x, y_2)| \le L|y_1 - y_2|$. Roughly, F must not change infinitely fast as y changes. Our equation is $y' = \pm 2\sqrt{y}$, and $\sqrt{y}$ has infinite slope at $y = 0$: it is not Lipschitz there, and that is exactly where uniqueness fails. (Giuseppe Peano showed in 1886 that continuity of F alone guarantees that a solution *exists*, but not that it is unique.) The proof of Picard–Lindelöf uses an iteration that converges by the contraction mapping principle of Section 2.2.

The leaking bucket

Here is a physical example. A bucket with a hole in the bottom empties according to Torricelli's law: the height h of water satisfies $h' = -k\sqrt{h}$. Run time forwards and everything is fine: the bucket empties and stays empty. But run time backwards: if you find an empty bucket, the equation cannot tell you when it emptied. The solution $h = 0$ for all time and the solutions that were full until some moment and then drained are all consistent with "empty now". The non-uniqueness is the mathematical shadow of the fact that information about the past has been lost.

Norton's dome

In 2003 the philosopher John Norton described a frictionless dome shaped so that a ball placed exactly at the top obeys Newton's laws in the form $r'' = \sqrt{r}$. One solution is

that the ball sits at the top for ever. But there are others, in which it sits there for any length of time and then spontaneously rolls off in any direction, without any cause. The example is still debated: is classical mechanics deterministic after all? Mathematically, it is non-uniqueness of exactly the kind in this question.

Try this next

1 **Two rates.** Try STEP II 2018, Question 8 (step.maths.org/questions/18-s2-q8). It solves two growth equations by substitution. Then ask why $y = 0$ is also a solution with the same starting value, and what that says about uniqueness.

2 **The leaking bucket.** Solve $h' = -k\sqrt{h}$ with $h(0) = h_0 > 0$. When is the bucket empty? Write down two different solutions with $h(T) = 0$ for some T and $h(t) = 0$ for all $t \geq T$.

3 **Uniqueness restored.** Show that $y' = 2y$ with $y(0) = 0$ has only the solution $y = 0$. (*Hint:* consider $\frac{\mathrm{d}}{\mathrm{d}x}(ye^{-2x})$.) What property does $2y$ have that $2\sqrt{y}$ lacks?

4 **Cubes.** Find all solutions of $y' = 3y^{2/3}$ through the origin.

Further reading

- **Morris Hirsch, Stephen Smale & Robert Devaney, *Differential Equations, Dynamical Systems, and an Introduction to Chaos*** (Academic Press). Existence and uniqueness, and much of the rest of this chapter.
- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). A classic, full of history; discusses singular solutions and envelopes.
- **John Norton, "The dome: an unexpectedly simple failure of determinism"**, *Philosophy of Science* 75 (2008). Short and readable.

The full guided solution is in Section 8.1:
https://beyondstep.keithmcnulty.org/ch08/non-uniqueness.html